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Effects of alloy additions on the microstructure and properties of Cr–Cr_2Nb alloys

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Abstract

Alloying additions of Ni, Co, Fe, Al and Re at levels up to 16 at.% were added to Cr–Cr_2Nb alloys containing 5.6–17 at.% Nb for the study of their microstructure, oxidation behavior, and mechanical properties. These alloys contain patches of primary Cr-rich solid solution surrounded by the eutectic structure having Cr_2Nb and Cr(~ 6% Nb) phases. The supersaturated Cr-rich solid solution in the cast alloys precipitated out secondary Cr_2Nb (Laves-phase) particles upon annealing at and above 900 °C. The studies by TEM and electron microprobe analysis indicated that the transition elements of Fe, Co, and Ni partition strongly in the Cr_2Nb-type Laves phase whereas rhenium and aluminum were only moderately enriched in the Laves phase. The Cr–Cr_2Nb alloys with $\leq$ 12% Nb exhibited considerable plastic deformation under compression tests. The yield strength of the alloys depended strongly on the volume fraction of the hard Laves phase. The oxidation resistance also increased with this volume fraction. Among the alloying additions, rhenium was the only element that substantially hardened the Cr–Cr_2Nb alloys at room temperature and 1000 °C. The hardening behavior is discussed in terms of partitioning and sublattice occupation of the alloying elements as well as altering the volume fracture of the hard Laves phase. Multilayer oxide products formed on these alloys upon exposure to air of an elevated temperature. Little beneficial effect of any of the alloying additions on oxidation behavior at 950 °C was found.

Keywords: Alloy addition; Cr–Cr_2Nb alloys; Microstructure; Mechanical properties

1. Introduction

The Laves-phase alloy, Cr_2Nb, with a C15 complex cubic crystal structure is of technological interest. This is because Cr_2Nb has a high-melting point (1770 °C) and relatively low density (7.7 g cm^{-3}). It has been reported that Cr_2Nb has excellent strength and creep resistance at elevated temperatures [1–4]. The major drawback for structural use of this alloy is that single-phase Cr_2Nb is very hard (~ 800 DPH) and brittle at room temperature [3]. Considerable effort has been devoted to understanding the phase transformation, deformation and fracture of Cr_2Nb and other refractory-type Laves phases [2–12].

The phase diagrams of the Cr–Nb, Cr–Zr, and Cr–Hf systems [13] show the formation of eutectic structures containing a solid solution Cr phase and a Cr_2X Laves phase, with the eutectic reaction occurring around 1600 °C. This phase relationship suggests the possibility of toughening the hard Cr_2X phase with a softer Cr-rich phase. Limited work on Cr–Cr_2Nb and Cr–Cr_2Hf indicates that the two-phase alloys have attractive properties for structural use [3–6,14–16]. For instance, the Cr-12 at.% Nb alloy has excellent strength at high temperatures and shows considerable compressive ductility at room temperature [3]. The microstructure of the two-phase alloys can be manipulated by various heat treatments. Preliminary efforts have shown that the precipitation of Cr_2Nb from the Cr-rich solid solution is sluggish, and particle growth is extremely slow in Cr–Nb alloys.

The Cr_2Nb phase, like many other Laves phases, has a wide range of compositional homogeneity [17], typically more than 5 at.% at elevated temperatures. This suggests the possibility of altering the mechanical properties of Cr_2Nb by alloy additions. At present, only limited results have been reported on alloying effects on Cr_2Nb [6]. In this study, alloy additions of Al, Ni, Co,

Fe and Re were added to the Cr–Nb alloys containing up to 12 at.% Nb. The partitioning of these elements was studied by electron microprobe analysis and transmission electron microscopy (TEM). The mechanical properties of the alloys were characterized by compression testing at room temperature and at 1000 °C, and the oxidation behavior of the alloys was studied by exposing specimens to air at approximately 950 °C.

2. Experimental procedures

Binary, ternary, and multi-element alloys based on Cr–Cr_2Nb were prepared by arc melting and drop casting into a copper mold (2.5 cm diam. x 10 cm long) preheated to 300 °C. Preliminary experiments showed little effect of impurities on the mechanical or oxidation behavior of the Cr–Cr_2Nb alloys. Nevertheless, in order to reduce the interstitial content in the alloys, high-purity Nb and Cr metal chips were normally used as charge materials, and the alloys were melted in a vacuum (10^{-3} Pa) furnace. For instance, the oxygen and carbon contents in Nb chips were less than 200 and 50 wt ppm, respectively; and the oxygen level in the alloys was typically less than 500 wt ppm. Table 1 lists the alloy compositions used in this study.

In order to control microstructures, alloy samples were sectioned from the ingots and heat treated at temperatures to 1200 °C for up to 3 days in vacuum. Compressive specimens with dimensions of 6 mm diam. x 9 mm long were machined by electrodischarge machining (EDM) after a final heat treatment. The specimens were tested in compression, using an Instron testing machine, at room temperature and 1000 °C at a nominal strain rate of 3.3×10^{-3} s^{-1}. The 1000 °C tests were carried out in vacuum ($\sim 10^{-4}$ Pa).

Table 1
Nominal compositions of Cr_2Nb–Cr (CN) alloys

Alloy number	Composition (at.%)
CN-7	Cr–6Nb
CN-4	Cr–12Nb
CN-45	Cr–17Nb
CN-46	Cr–12Nb–8Al
CN-47	Cr–12Nb–12Al
CN-48	Cr–12Nb–18Al
CN-49	Cr–12Nb–8Ni
CN-50	Cr–12Nb–12Ni
CN-43[a]	Cr–12Nb–4Re–2Al
CN-56	Cr–12Nb–10Co–4Re–2Al
CN-7A	Cr–6Nb–8Al
CN-52[b]	Cr–5.6Nb–4Re–1.5Al
CN-53[b]	Cr–5.6Nb–2Ni–2Fe–1.5Al

[a] Doped with 0.03 at.% Y.
[b] Doped with 0.1% Hf.

Microstructures of these alloys were examined by optical microscopy. Metallographic specimens were polished using a Syntron machine and then etched in a solution of 15 g KOH, 15 g $K_3Fe(CN)_6$, and 90 ml H_2O for approximately 5 s. An electron microprobe equipped with wavelength-dispersive spectrometry (WDS) was used to study the alloy composition and partitioning of elements between phases. Specimens used for the transmission electron microscopy (TEM) studies were cut into 3 mm disks by EDM and hand ground to a thickness of approximately 50 μm. Ion milling at 6 kV was used since the two-phase material could not be successfully electropolished. Specimens were examined in a Philips CM30 electron microscope at 300 kV. Energy dispersive spectroscopy (EDS) was performed in a Philips CM12 electron microscope with an EDAX 9100 spectrometer. The EDS analysis was made on precipitates that intersected ion milling perforations, thereby minimizing the contribution from the surrounding material.

Rectangular specimens (approximately 8 x 8–16 x 1 mm) were used for the oxidation studies. They were cut from the as-cast ingots by EDM. All major surfaces were mechanically ground with 600-grit SiC abrasive prior to oxidation. To evaluate any effect of heat treatment on the oxidation behavior, a few specimens were annealed in vacuum at 1100 °C for 3 days prior to exposure. Oxidation specimens were isothermally exposed to flowing, preheated, dried air (2 cc s^{-1}) at about 950 °C for up to 180 h. Weight changes were continuously measured using Cahn 1000 microbalances and a computer-controlled data acquisition system. At the end of each experiment, the specimens were furnace cooled and visually checked for loss of corrosion products by spallation. They were not re-exposed. In a few cases, specimens of selected compositions were held in laboratory air in a horizontal tube furnace at 950 °C and withdrawn at intervals of 24–72 h, cooled to room temperature, visually examined, weighed, and returned to the furnace. This latter type of experiment was used to qualitatively assess the effect of repeated damage from thermal cycling on the integrity of the corrosion product scales. Selected specimens from both types of exposures were analyzed by optical imaging and by scanning electron microscopy (SEM) and associated EDS.

3. Results

3.1. Structure and properties of binary Cr–Cr_2Nb alloys

The recently published Cr–Nb phase diagram indicates that the Cr–Cr_2Nb eutectic composition is 18.5

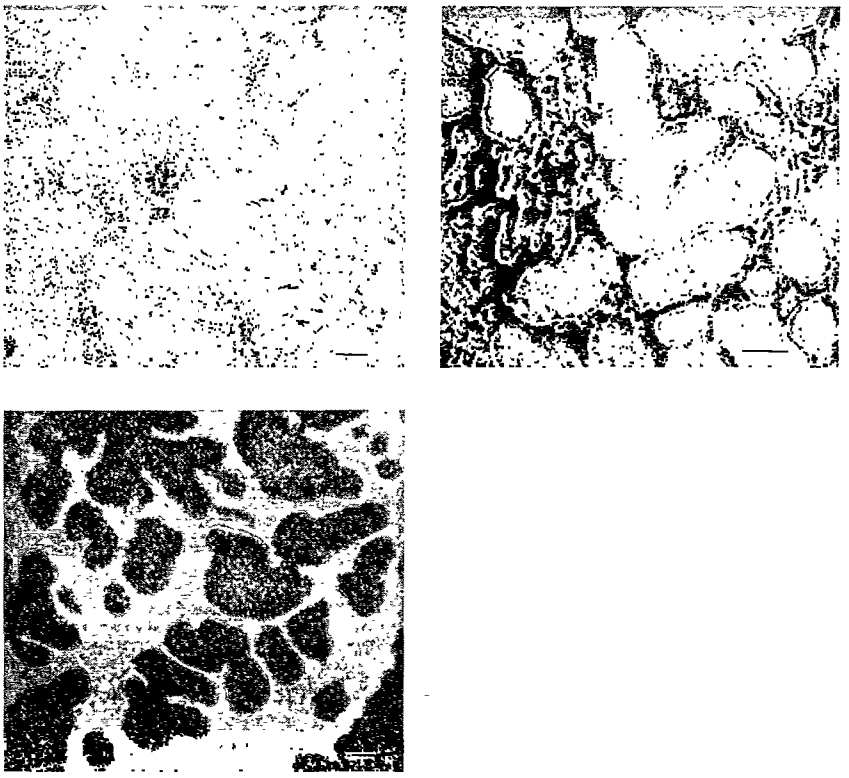


Fig. 1. Microstructures of Cr-12% Nb (CN-4): (a) optical micrograph of as-cast specimen, (b) optical micrograph of specimen annealed for 3 days at 1000 °C, and (c) backscattered electron image of specimen annealed for 3 days at 1000 °C (Bar is 10 μm).

at.% Nb [17]. As shown in Table 1, the three binary alloys containing 6, 12, and 17% Nb are therefore located in the hypoeutectic composition range and this has been confirmed by the present metallographic observations. The as-cast structure shows primary Cr-rich solid-solution patches which are surrounded by an interconnected Cr–Cr_2Nb eutectic structure Fig. 1(a). Because of the sluggish precipitation of Cr_2Nb from the Cr-rich phase, [3] no precipitates are visible (at least at the optical microscopy level) in the as-cast materials. The supersaturated Cr-rich solid solution precipitates out fine Cr_2Nb particles after annealing at 900–1200 °C, producing uniform distribution of extremely fine Cr_2Nb particles, see Fig. 1(b) and (c).

The mechanical properties of the three binary alloys were determined by compressive testing in air at room temperature. Table 2 summarizes the compressive properties of the binary alloys annealed for 3 days at 1100 °C prior to testing. With increasing niobium concentration (or vol. fraction of Cr_2Nb), the yield strength increases steadily and the ductility decreases accordingly. For the alloy with 17% Nb, the alloy specimens fractured prior to macroscopic yielding at room temperature, indicating brittleness of the alloy. Thus, for structural applications the niobium content in the alloys should be less than 17%. Since the maximum solubility of niobium in the Cr-rich phase is $\sim 5.6\%$, the coarse eutectic Cr_2Nb structure is expected to be avoided in the Cr–Nb alloys containing a smaller amount of niobium.

Table 2
Effect of niobium concentration on room-temperature compressive properties of Cr–Nb binary alloys

Alloy number	Nb concentration (at.%)	Strength (MPa)		Ductility (%)
		Yield	Ultimate	
CN-7	6	868	1820	11.4
CN-4	12	1140	1580	6.8
CN-45	17	1340	1340	~0

3.2. Microstructure and phase composition of Cr–Cr_2Nb with alloying additions

The ternary and multi-element alloys listed in Table 1 basically contain two phases, i.e., a Cr_2Nb-type Laves phase and a Cr-rich solid solution. In the as-cast condition, these alloys, just like the binary alloys, exhibit patches of the supersaturated Cr-rich solid-solution phase surrounded by an eutectic region. The supersaturated Cr-rich patches started to precipitate out fine Cr_2Nb particles at 900 °C, with particle coarsening at higher temperatures. Fig. 2 shows a backscattered electron image of CN-43 and CN-52 alloys annealed for 3 days at 1100 °C. The Laves phase with a higher Z number shows a bright contrast, whereas the Cr-rich

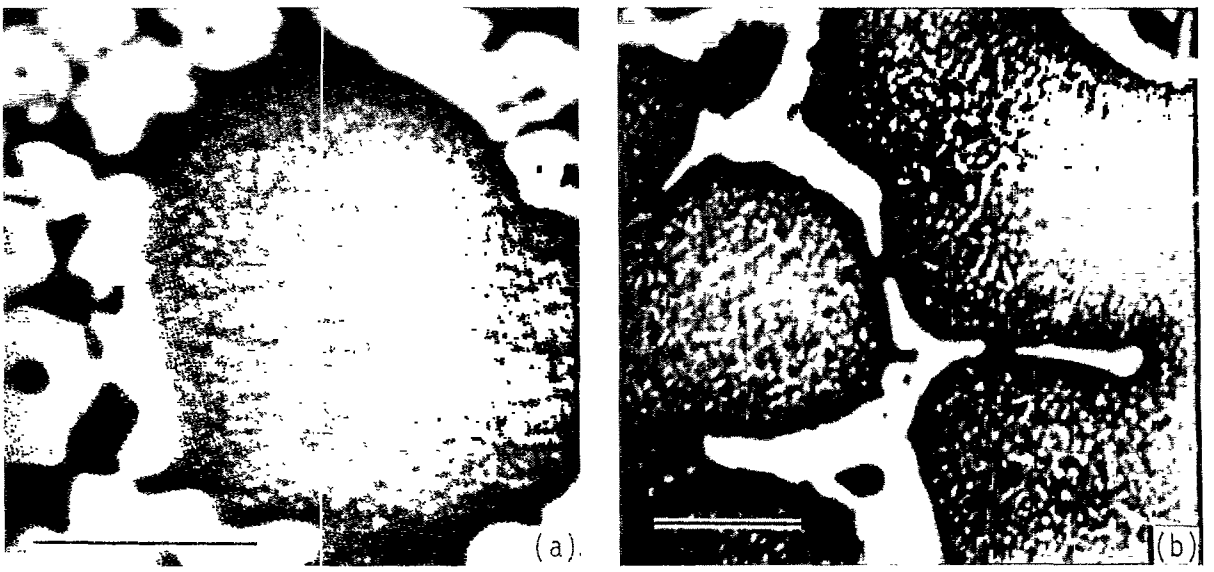


Fig. 2. Backscattered electron image of (a) CN-43 and (b) CN-52 specimens annealed for 3 days at 1100 °C (Bar is 10 μm).

solid solution with a lower Z number shows a dark gray contrast in Cr-rich patches, see Fig. 2. Here, fine Laves precipitates (in bright contrast), are clearly visible in Cr-rich patches seen in Fig. 2. Precipitate-free zones near the eutectic Laves phase are also observed. Precipitation of the Laves phase from the Cr-rich solid solution is apparently sluggish. These precipitates are extremely fine, with a size of ~ 0.3 μm for the alloys specimens annealed for 3 d at 1100 °C. Microporosity is also observed sometimes, with pore size typically less than 20 μm.

The compositions of the phases present in the alloys CN-43, -52, and -53 annealed for 3 days at 1100 °C were determined by electron microprobe analyses. Table 3 summarizes the results determined by WDS.

Table 3
WDS compositional analyses of the chromium–niobium alloys CN-43, -52 and -53 using electron microprobe

Alloy	Cr_2Nb-type phase (at.%)	Cr-rich phase (at.%)
CN-43	Cr = 64.5	92.9
	Nb = 30.1	3.4
	Re = 3.5	2.3
	Al = 1.9	1.4
CN-52	Cr = 63.7	94.1
	Nb = 32.6	2.9
	Re = 2.1	2.1
	Al = 1.6	0.9
	Hf = 0.01	0.03
CN-53	Cr = 55.1	96.6
	Nb = 29.1	1.0
	Fe = 5.2	1.2
	Ni = 8.2	0.6
	Al = 2.4	0.6

Note that the data reported in Table 3 represent an average of 3–5 analyses. The Cr-rich phase contained low levels of Nb, 2.9–3.4% in CN-43 and -52 and 1% in CN-53. Because of the fine and dense Cr_2Nb-type precipitates in the Cr-rich patches (see Fig. 2), the niobium contents reported here are probably higher than the actual value. The Cr_2Nb-type Laves phase, however, contained 30.3–32.6% Nb in CN-43 and -52, and 29.1% Nb in CN-53. Unexpectedly, iron and nickel in CN-53 partition strongly in the Cr_2Nb-type phase, instead of the Cr-rich solid solution. The partitioning ratio of nickel and iron in the two phases is 14:1 and 5:1, respectively. Aluminum and rhenium are only moderately enriched in the Laves phase.

The CN-43 alloy in the as-cast condition and in the annealed condition (3 days at 1200 °C) was examined by TEM. The as-cast specimens showed Cr-rich solid-solution patches surrounded by eutectic regions. The Laves phase in the eutectic region has a dark contrast in the TEM micrographs shown in Fig. 3(a). The bright areas are multiple perforations in the film resulting from ion milling. In the as-cast specimens, no precipitation of the Laves phase was noted. The Laves phase in the eutectic region was extensively internally twinned; an example is shown in Fig. 3(b). Such microtwinning has been observed in many Laves phases [7,9–11,14,15]. Dislocation pile-ups are observed in the Cr-rich phase impinging on the eutectic structure.

After an anneal for 3 days at 1200 °C, CN-43 specimens were found to have an extensive secondary precipitation of the Cr_2Nb-type Laves phase with a diameter of about 0.7 μm, Fig. 4(a). Not many dislocations were observed in this annealed material. These secondary precipitates were also intensely twinned, Fig. 4(b).

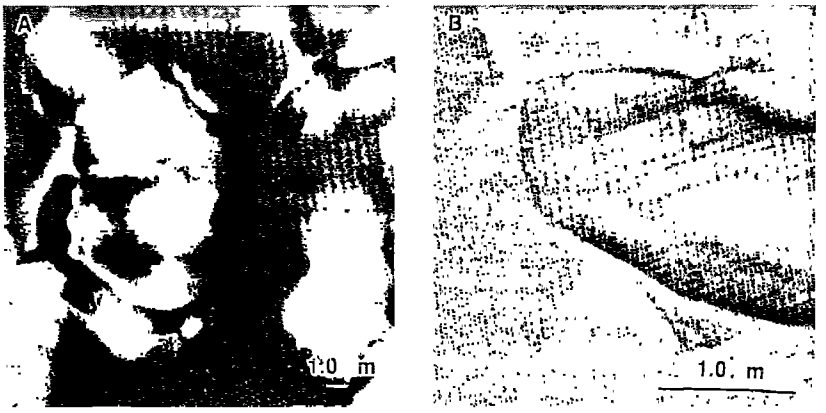


Fig. 3. TEM micrographs of as-cast CN-43 showing (a) chromium-rich patches surrounded by the eutectic structure and (b) a higher magnification micrograph showing the twinned structure in the Cr_2Nb phase and the dislocation pileups next to the Cr_2Nb phase.

The results of EDS analysis of the two CN-43 specimens examined by TEM are given in Table 4. The spectra were collected from the intersection of each phase with the ion-milling perforations resulting in little interference from the neighboring phase. In the as-cast material, the dark phase in the eutectic structure is close to the expected composition of Cr_2Nb. The rhenium and aluminum additions are fairly evenly distributed between the Cr-rich solid solution and the Laves phase. The niobium content in the Cr-rich phase was determined to be about 5.5%. After an anneal for 3 days at 1200 °C, both the coarse Cr_2Nb-type phase in the eutectic structure and the secondary precipitates in the Cr-rich phase are moderately enriched in both aluminum and rhenium. Furthermore, most of the niobium has precipitated out as secondary precipitates, with about 1% Nb remaining in the Cr-rich solid solution. Also, the solid-solution phase contains as much as 95% Cr. All these results agree quite well with those from the electron microprobe analyses.

3.3. Mechanical properties of $Cr–Cr_2Nb$ with alloying additions

The mechanical properties of the Cr-12% Nb and Cr-7% Nb alloys with alloying additions were determined by compression testing at room temperature in air and at 1000 °C in vacuum. All specimens were annealed for 3 days at 1100 °C prior to testing. Table 5 summarizes the compressive properties of the alloys containing Al, Ni, Co, Fe, and Re additions. At room temperature, the alloying elements, with the exception of rhenium, do not strongly affect the yield strength and ductility of the Cr–Nb alloys. Rhenium is the only element that appears to significantly increase the yield strength of the Cr-12% Nb alloy. At 1000 °C, rhenium substantially increases the yield strength of the Cr-12% Nb and Cr-7% Nb alloys whereas other alloying elements lower the strength. CN-53, containing 2% Fe, exhibited a distinctly lower yield strength at 1000 °C as compared with other alloys containing no iron. This suggests that alloying with iron softens the Cr–Nb alloys at elevated temperatures. The ductility at 1000 °C appears to be not much affected by alloy additions.

3.4. Oxidation behavior

Fig. 5 shows the effects of compositional modifications to Cr-12% Nb on isothermal oxidation kinetics in air at approximately 950 °C. Neither the addition of Al nor Re proved to be beneficial. Indeed, the binary alloy showed better isothermal oxidation behavior than any of the more highly alloyed compositions. Approximate parabolic rate constants (fitted to the data for times greater than 40 h ranged from $(5–16) \times 10^{-5}$ $mg^2\ cm^{-4}$ s for these alloys compared to 2×10^{-5} $mg^2\ (cm)^{-4}$ s for the binary composition. Visual examination of the specimen surfaces after cooling from the exposure temperature showed that, with the exception of CN-4 and -48, the 12% Nb compositions suffered significant scale spallation. For example, while about 50–80% of the oxidized surfaces of many of the more highly alloyed compositions showed evidence of spallation after cooling from 950 °C, less than 10% of the scale formed on CN-4 was lost.

The isothermal gravimetric results for the Cr-6% Nb alloys are shown in Fig. 6, which includes data for as-cast (ac) and heat-treated (ht) specimens of CN-52 and -53. Alloying with 8% Al or 4% Re did not significantly improve isothermal oxidation resistance relative to the binary composition; upon cooling from the exposure temperature, the spallation losses for CN-7A and CN-52 were comparable and these latter two alloys showed higher weight gains and nonparabolic (linear) reaction kinetics. Although CN-53 did show the lowest weight gains, the long-term parabolic rate constants (k_p's) were similar to that for CN-7 and the difference in oxidation behavior is not considered substantial. The influence of heat treatment on oxidation behavior can be assessed by comparing ac to ht results

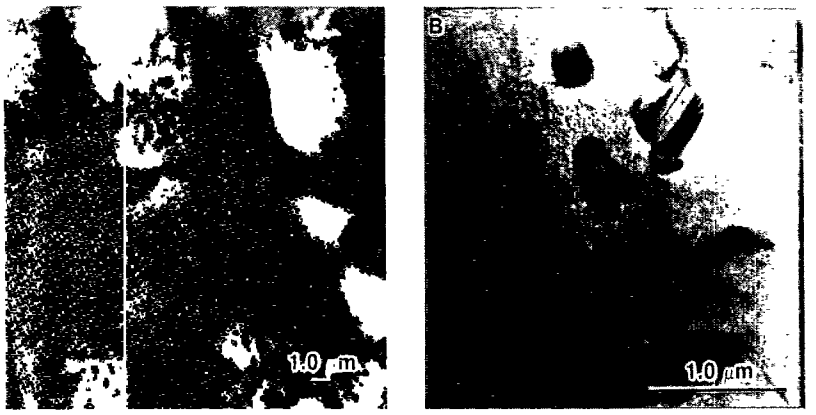


Fig. 4. TEM micrographs of CN-43 after an anneal for 3 days at 1200 °C showing (a) Cr_2Nb precipitates in the chromium-rich patches surrounded by the eutectic structure and (b) a higher magnification micrograph of the internal twinning in Cr_2Nb precipitates.

for CN-52 and -53 (Fig. 6). The differences in overall isothermal oxidation behavior were small, with the ac specimens showing slightly higher weight gains, but approximately equal k_p's, relative to the ht versions of the same composition. Therefore, in general, such variations in starting microstructure did not appear to change the relative effects of the observed influences of compositional modifications on longer-term ($> \sim 30$ h) oxidation behavior at 950 °C.

The addition of small amounts of active elements to many chromia-forming alloys can significantly reduce the oxidation rates and improve scale adherence (see, for example [18–20]). In the present case, however, neither yttrium added to Cr-12% Nb nor hafnium in Cr-6% Nb appear to proffer such benefits. However, the concentration of these elements may have been too low to be effective in this regard for these particular alloys. Furthermore, the compositions were not designed to directly study the effects of these active elements on oxidation behavior.

As described above, and as also shown in previous work, [21] the corrosion products grown on isothermally oxidized binary Cr-12% Nb had reasonably good adherence to the alloy substrate, particularly with respect to the other $Cr–Cr_2Nb$ alloys under investigation. It therefore was of interest to examine the behavior of Cr-12% Nb under the more aggressive conditions imposed by the repeated application of stresses generated during cooling from, and heating to, the oxidation temperature (950 °C) and to evaluate possible effects of alloying additions under such conditions. Fig. 7 compares cyclic oxidation data for Cr-12% Nb with the isothermal results reported above; each datum point for the cycled specimens represents the weight change after an individual thermal cycle. Note that thermal cycling

Table 4
EDS/TEM compositional analyses of CN-43 (at.%)

	Al	Nb	Cr	Re
As Cast				
Cr-rich phase	4	5.5	85.5	5
Large Cr_2Nb phase	5.5	29	62	3.5
Annealed				
Cr-rich phase	2	1	95	2
Large Cr_2Nb phase	3.5	28	63.5	5
Small precipitates	6	29	60	5
Nominal (plus 0.04 wt.% Y)	2	12	82	4

Table 5
Compressive properties of $Cr-Cr_2Nb$ alloys

Alloy number	Composition (at.%)	Strength (MPa) Yield	Ultimate	Strain (%)
Room temperature				
CN-4	Cr–12Nb	1140	1750	6.8
CN-46	Cr–12Nb–8Al	1090	1540	4.7
CN-49	Cr–12Nb–8Ni	1110	1610	6.8
CN-50	Cr–12Nb–12Ni	1250	1740	6.1
CN-43	Cr–12Nb–4Re–2Al	1490	2190	5.8
CN-56	Cr–12Nb–10Co–4Re –2Al	1460	1870	5.5
CN-7	Cr–6Nb	868	1820	11.4
CN-52	Cr–5.6Nb–4Re–1.5Al	850	1670	>15.0[a]
CN-53	Cr–5.6Nb–2Ni–2Fe –1.5Al	749	1660	>12.0[a]
1000°C				
CN-4	Cr–12Nb	685	855	22.0
CN-46	Cr–12Nb–8Al	616	980	14.0
CN-49	Cr–12Nb–8Ni	601	682	16.0
CN-50	Cr–12Nb–12Ni	513	640	17.0
CN-43	Cr–12Nb–4Re–2Al	877	1070	20.0
CN-56	Cr–12Nb–10Co–4RE –2Al	527	947	>6.0[a]
CN-7	Cr–6Nb	468	702	31.0
CN-52	Cr–5.6Nb–4Re–1.5Al	548	1190	28.0
CN-53	Cr–5.6Nb–2Ni–2Fe –1.5Al	313	738	26.0

[a] The tests were stopped at the indicated strain.

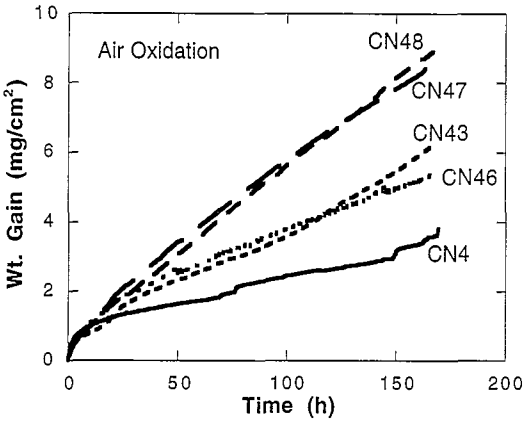


Fig. 5. Weight change versus time for Cr-12% Nb alloys isothermally oxidized in air at 950–965 °C.

increased overall specimen weight gains and spallation losses relative to the specimen that was exposed under isothermal conditions. The addition of 8% Al did not significantly change the cyclic oxidation resistance of the Cr-12% Nb alloy relative to what is shown in Fig. 7. In both cases, numerous cracks, voids, and delaminations in the scales were observed. An example of such is shown in Fig. 8, which is a polished cross section of a CN-4 specimen that had been exposed at 950 °C for seven thermal cycles totaling 268 h of exposure. The additional scale damage caused by repeated thermal cycles is also shown in Fig. 9, which compares SEM micrographs of CN-4 specimens that had experienced one and seven cycles. The thermal stresses from several heating and cooling cycles degraded the ability of the corrosion product layer to protect the underlying alloy and resulted in the substantially higher weight changes for the cyclically oxidized specimens (Fig. 7). Preliminary cyclic oxidation results for CN-43 showed that it gained weight at a faster rate than the binary Cr-12% Nb alloy, which is consistent with the isothermal results shown in Fig. 5.

The examination of cross sections and damaged scales (see Fig. 10) revealed that multilayer oxidation

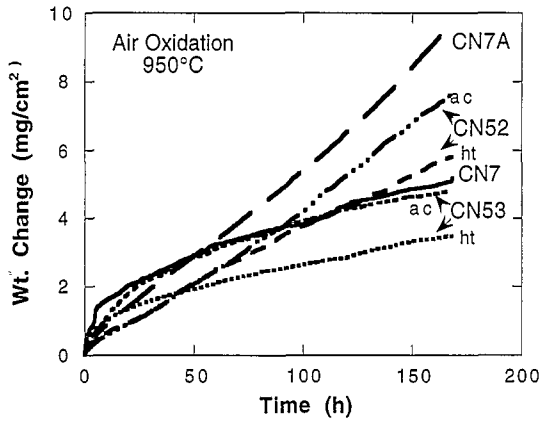


Fig. 6. Weight change versus time for Cr-6% Nb alloys isothermally oxidized in air at 950 °C.

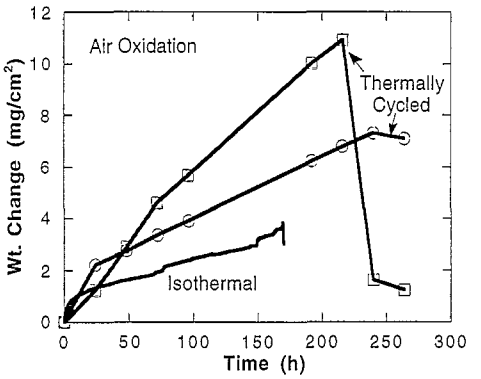


Fig. 7. Weight change versus time for two binary Cr-12% Nb specimens cyclically oxidized in air at 950 °C

products had formed on the Cr–Cr_2Nb alloys. Analysis by EDS showed that the upper layer consisted primarily of chromium cations, while underlying layers contained both chromium and niobium (and some aluminum for those alloys containing this element). These results are consistent with the microstructural analysis previously conducted on oxidized binary Cr–Nb alloys [21]. This prior work also showed that a distinct thin, porous layer of mixed (Cr, Nb) oxide preferentially formed on the Cr-rich phase.

4. Discussion

The phase diagram of the Cr–Nb system [17] indicates the formation of two eutectic systems, i.e., Cr–Cr_2Nb and Nb–Cr_2Nb, with the eutectic temperatures above 1600 °C. Both eutectic systems have the potential to be developed as new structural materials, and at present, considerable effort is being devoted to these alloys [2–7,14–16]. The advantage of the Nb–Cr_2Nb eutectic system is that the Nb-rich solid solution is more ductile than the Cr-rich solid solution at ambient temperatures. However, the Nb-rich solid solution has poor oxidation resistance below the temperature where chromia has significant volatility [2] (i.e., < 1100 °C). It is also known that Nb- and Ta-base alloys, even when ductile initially, can be easily embrittled when contaminated with interstitial impurities during service in hostile environments [22,23].

The Cr–Cr_2Nb alloys studied here contain primary Cr-rich solid-solution patches surrounded by an eutec-

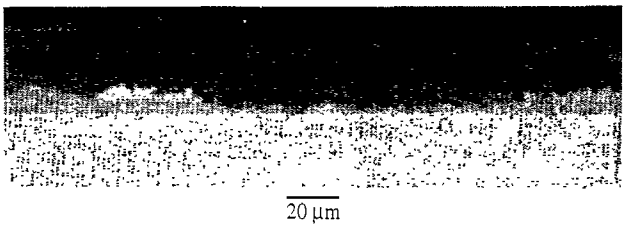


Fig. 8. Polished cross section of Cr-12% Nb-8% Al (CN-4) cyclically oxidized in air at 950 °C for 268

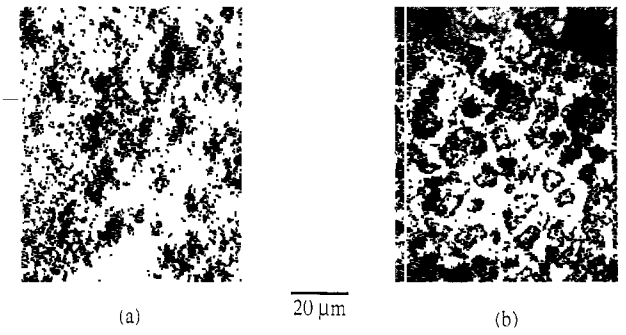


Fig. 9. Scanning electron micrographs of Cr-12% Nb after oxidation in air at 950 °C (a) after cooling from isothermal exposures for 168 h. (b) after seven thermal cycles totaling 268 h of exposure

tic structure containing Cr_2Nb and Cr (~6% Nb) phases. Because of the sluggish nature of the precipitation reaction, i.e., Cr (5.6% Nb) → Cr_2Nb + Cr (~1% Nb), it is found that the supersaturated Cr-rich solid solution is stable to room temperature in the cast alloys. Fine and dense Cr_2Nb particles are precipitated out of the solution when annealing at 900 °C and above. Cr_2Nb precipitates are also found to grow slowly even above 1000 °C, possibly due to the slow diffusion of niobium in the Cr-rich solid solution as well as the low interfacial energy of Cr_2Nb particles.

Alloying additions of Al, Re, Fe, Co and Ni at levels up to 16% were added to Cr–Cr_2Nb alloys. The selection of these alloying elements was based on attempts to improve oxidation resistance (aluminum) and ductilize the Cr-rich solid solution. Rhenium is known to be effective in ductilizing the group VIA elements (i.e., Cr, Mo and W) [24]. The studies by TEM and electron microprobe analyses indicate that rhenium and aluminum are only moderately enriched in the Laves phase. The transition elements of Fe, Co, and Ni can be dissolved extensively in Cr-materials. It is unexpected to find that these transition elements partition strongly in the Cr_2Nb Laves phases, instead of the Cr-rich solid solution.

The stoichiometric composition of Cr_2Nb has 33.3 at.% Nb. When an alloying element is added to the Laves phase, it may occupy the chromium site, the Nb site, or both sublattice sites. A careful examination of the composition of the Cr_2Nb-type Laves phase in



Fig. 10. Scanning electron micrograph of cyclically oxidized surface of Cr-6% Nb showing multilayer corrosion product layers.

Table 6
Atomic size and electron concentration (e/a) of alloying elements together with stability of X_2Nb Laves phase

Element	Goldschmidt[a] radii	e/a	X_2Nb[b]
Nb[a]	1.47	5	–
Al	1.43	3	None
Re	1.37	7	None
Cr[a]	1.28	6	Cr_2Nb
Fe	1.27	8	Fe_2Nb
Co	1.26	9	Co_2Nb
Ni	1.24	10	None

[a] Ref. [25].
[b] X = alloying element. No Cr_2X phases exist.

CN-43 and CN-52 (Table 3) indicates that rhenium and aluminum possibly occupy both the sublattice sites. However, the analysis of solid-solution hardening, as discussed later, suggests that rhenium may essentially occupy the niobium sites, instead of the chromium sites, in the Laves phase in CN-43 and -52. The measured composition of the Cr_2Nb-type phase in CN-53 indicates that iron and nickel occupy mainly the chromium sites. These results agree well with the study of solubility lobes in refractory intermetallics containing ternary additions [6].

In order to rationalize the partitioning behavior of alloying elements, it is necessary to consider the factors that control the stability of Laves phases such as Cr_2Nb. Laves phases are compounds whose stability is strongly associated with the atomic-size difference between their constituent elements [25,26]. It is also known that the electron concentration, (e/a), also plays an important role in determining the crystal structure of these compounds [26]. Table 6 lists the atomic size and e/a of the alloying elements used in this study, together with the stability of X_2Nb compounds, where X represents an alloying element. The partitioning behavior of the alloying elements shown in Tables 3 and 4 appears to be correlated best with the Goldschmidt radii, instead of e/a or the stability of X_2Nb. For instance, nickel atoms partition strongly in the Cr_2Nb phase, but no such Ni_2Nb Laves phase is reported to exist. The atom size of Fe, Co, and Ni is close to, but smaller than, that of chromium. These elements have a strong tendency to partition in the Cr_2Nb phase and to occupy the chromium sublattice sites. Nickel, which has the smallest atomic size, has the strongest tendency to partition in the Cr_2Nb phase. Rhenium and aluminum, whose atomic size falls between that of chromium and niobium, have a lower tendency to partition in the Cr_2Nb phase as compared with nickel and iron.

The yield strength of the Cr–Cr_2Nb alloys depends strongly on the volume fraction of the hard phase Cr_2Nb which is, in turn, proportional to the niobium

concentration in the alloys. Consequently, a linear relationship between the yield strength and the niobium is expected and was observed (Fig. 11).

Alloy additions mainly contribute to the yield strength of the Cr–Cr_2Nb alloys in two ways: (1) by increasing the volume fraction of the hard Laves phase and (2) by solid-solution hardening of the Cr-rich phase. Nickel and iron (presumably cobalt also) preferentially partition in the Laves phase, and thus contribute little to the solid solution hardening of the Cr-rich phase. Also, since nickel and iron occupy mainly the chromium subsite, they should not cause much increase in the volume fraction of the Laves phase. As a result, iron and nickel, probably cobalt as well, contribute little hardening in the Cr–Cr_2Nb alloys at room temperature. Rhenium is the only element that substantially hardens the Cr–Cr_2Nb alloys. This element may contribute some to the solid-solution hardening of the Cr-rich phase, but the major contribution may come from its preferred occupation of the niobium sublattice sites, resulting in the increase in the volume fraction of the Laves phase. As indicated in Table 4, precipitation of the Laves phase during annealing for 3 days at 1100 °C reduces the rhenium content in the Cr-rich phase from 5 to 2%. At 1000 °C, the yield strength of the alloys increases substantially with rhenium addition and decreases with iron. The addition of 5.2% Fe dissolved in the Laves phase may lower the strength of the phase at 1000 °C, resulting in decreasing the overall strength of CN-53.

Typically, multilayer scales on A–B alloys (where A is the more noble element and B is sufficiently concentrated) consist of BO or (A,B)O underlying AO or a mixed oxide [27,28]. In the present case, where chromium is slightly more noble than the oxides of niobium [29,30] this predicted layer structure is observed only in the broadest sense: Nb-containing products are always formed under a pure Cr_2O_3, but a

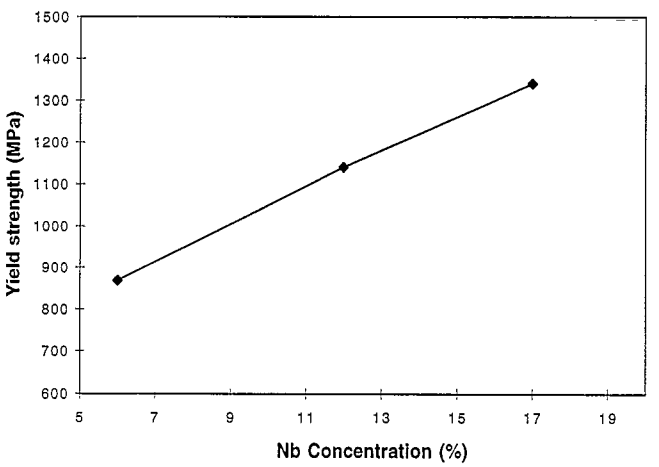


Fig. 11. Plot of the room-temperature yield strength as a function of niobium concentration.

product with very distinct ordered layers does not develop. A thin oxidation product preferentially forms directly on the Cr-rich phase such that a contiguous inner layer over the alloy surface is not observed [21]. It is possible that the phases at first oxidize independently of each other and, at a later stage, form a uniform scale in the plane of the surface. However, more detailed analyses of scale products, particularly for short oxidation times, are needed before any definitive conclusions regarding scale growth mechanisms can be made.

The binary Cr-12% Nb alloy had better oxidation resistance than one with 6% Nb, both in terms of growth kinetics (compare the gravimetric data for CN-4 in Fig. 5 with that for CN-7 in Fig. 6) and scale adherence [21]. Because the increase in niobium concentration from 6–12% results in a higher volume fraction of the eutectic mixture, the poorer overall oxidation resistance of the 6% Nb alloy appeared to be related to the greater abundance of the Cr-rich phase [21]. This hypothesis regarding the importance of the amount of Cr-rich phase in determining overall oxidation resistance of the Cr–Cr_2Nb alloys is consistent with both the present and previous results showing some unique similarities with the growth, cracking, and spallation behavior of scales grown on chromium [18,21,28,31,32]. Consequently, as most of the alloying additions shown in Table 1 either preferentially partition to the eutectic phase (particularly aluminum) or are reasonably evenly distributed, little beneficial effect of these elements on the overall oxidation resistance was found.

The chromium-like aspects of the scaling behavior of Cr–Cr_2Nb would suggest that some of the proven methods of reducing the oxidation rate and improving the scale adherence of chromia-formers could be effective for this alloy system. However, as described above, initial attempts at microalloying with active elements (albeit at levels that may have been too low for this alloy system) were not effective in improving oxidation behavior.

The overall oxidation losses reported above are significant, but for most of the alloys, relatively moderate. The measured parabolic rates for the Cr-12% Nb alloys are well within the rather broad range reported for chromia growth [28]. There are numerous high-strength, high-temperature intermetallic alloys that show poorer oxidation resistance. For example, the total weight changes for the more resistant compositions oxidized at 950 °C are lower by about an order of magnitude than those for Nb–Cr_2Nb oxidized at 1000 °C [2]. However, as chromia plays an important role in the oxidation of the present Cr–Cr_2Nb alloys, higher temperature exposures will lead to accelerated oxidation as the oxide product is volatilized. This has been observed for the binary Cr-12% Nb alloy at 1100 °C, [33] but can be ameliorated by the use of appropriate high-temperature coatings [34,36].

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